School of Mechanical Engineering-
Mechatronics Department

AUTOMOTIVE ELECTRONICS

AUTO-SWITCHING FRONT LIGHT

for Motorcycles & Scooters

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Adaptive Front Lighting System

High Beam → Auto-Switch → Low Beam

LDR

light sensor

AFS

adaptive lighting

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Why Night Lighting Matters

50%

of fatal road crashes occur at night

WHO Road Safety 2023

3×

higher fatality risk driving after dark

NHTSA Research

60%+

of two-wheeler deaths happen after sunset

MoRTH India 2023

1.19M

road deaths globally per year (WHO 2023)

WHO 2023

🌙 Visibility Reduction

Human eyes adapt poorly to high-contrast light. A sudden high-beam can cause 5–8 seconds of temporary blindness enough to cause a fatal crash at 60 km/h.

🏍 Two-Wheeler Vulnerability

Motorcyclists are 28 times more likely to die per mile travelled than car occupants. They have no protective cabin glare is a direct, unmitigated danger.

🔦 The Glare Problem

Headlight glare from oncoming vehicles causes 'dazzle' that impairs vision. In India alone, over 4,500 deaths per year are attributed to glare-related accidents (MoRTH 2022).

How the Industry Handles Lighting Today

🏍 AHO Always-On Headlights (Two-Wheelers)

What it does
Automatic Headlight On (AHO) keeps headlights on at all times the engine is running there is NO manual override. First mandated in India in April 2017 for new vehicles.

How widespread
Most two-wheeler manufacturers now implement AHO as standard. In India, over 21 million two-wheelers are sold annually virtually all now include AHO.

Limitation
AHO keeps a fixed beam ON it does NOT adapt to oncoming traffic. High beam vs. low beam is still entirely the rider's manual responsibility.

Safety impact
Studies show AHO has reduced day-time two-wheeler accidents by ~10–15% by improving daytime visibility to other road users (IIHS 2022).

🚗 AFS Adaptive Front Lighting (Cars)

What it does
AFS automatically switches low beams based on vehicle speed and steering it adjusts lighting direction to compensate for road curvature.

Winding roads
On winding roads, AFS changes the lighting pattern to match road curvature, improving night visibility in bends where fixed beams fail.

Crossroads
At junctions, AFS illuminates the chosen direction of travel left or right giving the driver a preview of what is ahead in the turn.

Market status
AFS is standard on premium cars. The global automotive lighting market was $29.7B in 2022 and is forecast to reach $47.2B by 2028 (MarketsandMarkets 2023).

⚠ Critical Gap: AFS exists for cars — but NO equivalent system exists for motorcycles and scooters, which make up 70%+ of India's road traffic.

The Glare Hazard — Unaddressed for Two-Wheelers

The Core Problem

Oncoming high-beam glare
When a vehicle approaches with high-beam headlights, the driver experiences a blinding dazzle. This 'dazzle effect' can cause 5–8 seconds of temporary blindness.

No AFS for motorcycles
Unlike cars, motorcycles have no adaptive front lighting system. Riders must manually switch beams a dangerous distraction at speed or on bends.

Human reaction gap
The average human reaction time is 1.5 seconds. At 60 km/h, that means 25 metres covered blind. With a 5-second dazzle, that's 83 metres with zero visibility.

Frequency of exposure
In high-density traffic countries like India, a night rider encounters oncoming high-beam vehicles every 30–60 seconds on average — making manual switching impractical.

83m Blind distance at 60 km/h during dazzle

4,500+ Annual glare-related deaths in India (MoRTH)

60% Two-wheeler share of road accidents

Auto-Switching AFS for Motorcycles

The Core Idea

Implement an Adaptive Front Lighting System specifically for motorcycles and scooters. Using a Light Dependent Resistor (LDR) sensor, the system automatically detects oncoming vehicle headlights and switches from high beam to low beam without any rider action required.

OBJECTIVE 1

Detect Oncoming Headlights

An LDR sensor continuously monitors ambient light levels. When an oncoming vehicle's headlight is detected, the signal triggers the switching circuit automatically.

OBJECTIVE 2

Auto High→Low Beam Switch

Upon detection, a relay-controlled circuit switches the headlight from high beam to low beam instantly eliminating glare to the oncoming driver without rider input.

OBJECTIVE 3

Restore High Beam Post-Pass

Once the oncoming vehicle has passed and LDR ambient light drops back to baseline, the system automatically restores high beam for maximum forward visibility.

Circuit Block Diagram & System Flow

Oncoming Light Source

High-beam detection

→

LDR Sensor

Light intensity measurement

→

Comparator Circuit

Signal threshold processing

→

Relay / Switching

Beam mode control

→

Headlight Output

High / Low beam switch

🔋 Battery / Power Supply

Powers the entire system. DC supply feeds: LDR sensor circuit, comparator/op-amp, relay coil, and vehicle headlight assembly.

Step-by-Step Operation

01 🌙 **Night Riding Mode**

Rider is on high-beam mode during normal night driving. The LDR sensor records ambient darkness as baseline (low resistance = low light = high beam is active).

02 🔦 **Oncoming Light Detected**

An oncoming vehicle's headlight falls on the LDR sensor. Resistance increases as light intensity rises. The voltage output from the LDR circuit changes.

03 ⚡ **Comparator Triggers**

The comparator (op-amp) circuit detects that the LDR voltage has crossed a set threshold. It sends a HIGH signal to the relay control circuit.

04 🔌 **Relay Switches Beam**

The relay de-energises the high-beam circuit and activates the low-beam circuit. This happens in milliseconds faster than any human reflex.

05 ✅ **Auto High-Beam Restore**

Once the oncoming vehicle passes, LDR light drops. The comparator resets, relay switches back, and high beam is automatically restored for full forward vision.

Hardware Components Used

Resistance range: 1–10 MΩ (dark) to ~100 Ω (bright)

LDR (Light Dependent Resistor)

Photoresistor whose resistance decreases as light intensity increases. Acts as the primary sensor placed facing oncoming traffic. Low cost: ~₹5–20 per unit.

LM741 / LM393 — dual supply ±12V or single 5V

Op-Amp / Comparator IC

Compares LDR output voltage against a reference. When LDR voltage exceeds the threshold (bright incoming light), comparator output goes HIGH to trigger relay.

5V / 12V relay coil — SPDT or DPDT configuration

Relay Module

Electromagnetic switch that physically connects or disconnects the high/low beam circuit. Relay allows low-current sensor signal to control high-current headlight load.

H4 dual-filament halogen / LED bulb (high + low beam)

Headlight Assembly

Standard H4 bulbs have two filaments — one for high beam, one for low beam. The relay switches between them. Works with both halogen and LED headlight types.

12V motorcycle battery (Li-ion or lead-acid)

Power Supply / Battery

Vehicle's existing 12V battery powers the entire system. Total additional power draw is minimal (<0.5W for sensor+comparator), adding no strain to the alternator.

The Two-Wheeler Lighting Market

21M+

units/year

Two-wheelers sold in India annually

SIAM 2023

$29.7B

Global automotive lighting market 2022

MarketsandMarkets 2023

$47.2B

by 2028

Projected market size — 8.1% CAGR

MarketsandMarkets 2023

70%

Share of India's vehicles that are two-wheelers

MoRTH 2023

🏭 Manufacturing Context

Honda, Bajaj, Hero MotoCorp collectively produce 15M+ two-wheelers annually. Adding an LDR-based AFS as standard equipment would add <₹200 to unit cost — negligible vs. retail price of ₹50,000–₹200,000.

🌍 Global Mandate Trend

The EU mandated daytime running lights for motorcycles in 2016. AHO is mandatory in India since 2017. Progressive headlight regulation is tightening — AFS for two-wheelers is the logical next step.

⚡ Technology Readiness

LDR sensors, op-amps and relay modules are mature, mass-produced components. The technology needed for this AFS is already in volume production for other applications — enabling low-cost retrofit kits.

Why India Is the Critical Market

21M

Two-wheelers sold per year

SIAM 2023

210M+

Registered two-wheelers on road

MoRTH 2023

1.55L

Road deaths in India 2022

MoRTH 2022

44%

Death share: two-wheelers

MoRTH 2022

60%+

Night accident proportion

WHO Study

₹200

Estimated AFS add-on cost

Component Analysis

Manual vs. Auto-Switch — What Changes?

❌ WITHOUT Auto-Switching

- ✗ Rider must manually switch from high to low beam
- ✗ Requires split attention dangerous at speed or on bends
- ✗ Average human reaction time: 1.5 second's delay
- ✗ Rider may forget to switch persistent glare hazard
- ✗ Multiple switches per km in heavy night traffic
- ✗ Fatigue from repeated manual interaction increases error risk
- ✗ Oncoming driver suffers 5–8 second temporary blindness

VS

✅ WITH Auto-Switching (AFS)

- ✓ LDR sensor detects oncoming beam automatically
- ✓ Zero rider input switches in milliseconds
- ✓ Reaction time: <50ms (100× faster than human)
- ✓ Consistent no forgetting, no fatigue factor
- ✓ Handles unlimited switches per km effortlessly
- ✓ Rider maintains full focus on the road ahead
- ✓ Oncoming driver protected from glare both safer

The Technology Gap

Feature	Car AFS	Motorcycle (Existing)	CDRA — Proposed
Auto High→Low Switch	✅ Standard	❌ Manual only	✅ **Via LDR circuit**
Oncoming detection	✅ Camera/radar	❌ None	✅ **LDR photosensor**
Road curvature adapt	✅ Steering angle	❌ None	🔄 **Future scope**
Response time	✅ <50ms	❌ ~1.5s human	✅ **<50ms relay**
System cost (add-on)	High ($200–$500)	₹0 (not avail.)	✅ **<₹200 (low-cost)**
Retrofit to existing	❌ Factory only	N/A	✅ **Yes — universal**
Two-wheeler availability	❌ Not available	❌ Not available	✅ **This project**

CDRA delivers the same core functionality as a car AFS at a fraction of the cost — specifically designed for the most vulnerable road users.

Why This Matters — Lives & Economics

🛡️ Accident Reduction Potential

AHO (always-on headlights) reduced daytime accidents by 10–15%. An auto-switching AFS targeting the more severe night-glare problem could realistically prevent thousands of annual fatalities in India alone.

5–15%
Estimated accident reduction potential

😌 Rider Focus & Fatigue

Eliminating repetitive manual beam switching reduces cognitive load on the rider particularly important on long highway rides where fatigue is a major accident contributing factor.

₹7,000Cr
Economic saving if 5% reduction achieved

⚖️ Regulatory Compliance Path

With India mandating AHO since 2017 and the EU mandating DRL since 2016, an AFS mandate for motorcycles is a logical regulatory progression. Early adoption creates market advantage.

<50ms
System response time vs 1.5s human

💰 Economic Impact

Road accidents cost India ~₹1.47 lakh crore ($18B) annually in economic losses (MoRTH 2022). Reducing night two-wheeler accidents by even 5% would save ~₹7,000 crore in economic impact.

<₹200
Add-on cost per vehicle (kit price)

Next Steps & Technology Roadmap

ADAS / ML classification

F1 — Camera-Based Detection

Replace LDR with a small CMOS camera + ML model to distinguish oncoming headlights from streetlights, reflections, or stationary lights dramatically reducing false positives.

Distance + velocity data

F2 — LiDAR / Radar Integration

Add distance sensing to predict time-to-pass activating low beam earlier on fast-approaching vehicles and restoring high beam sooner after they pass.

Neural network beam shaping

F3 — AI Adaptive Beam Control

Deep learning models could continuously optimize beam angle and intensity based on road type, speed, weather, and traffic density beyond simple on/off switching.

Vehicle-to-Vehicle protocol

F4 — V2V Communication

Connected vehicles could signal beam-switch requests to each other directly eliminating even the sensor processing delay for maximum responsiveness.

App-based configuration

F5 — Smartphone Integration

A companion app could allow riders to configure sensitivity thresholds, view system status and diagnostics, and receive maintenance alerts for the LDR and relay components.

Factory-standard across all OEMs

F6 — OEM Integration Mandate

Advocate with MORTH / ARAI for inclusion of auto-switching AFS as a mandatory safety standard for all new two-wheelers similar to AHO mandate of 2017.

Key Takeaways

This project presents a practical, low-cost Adaptive Front Lighting System for motorcycles and scooters using an LDR sensor and relay circuit to automatically switch between high and low beam when oncoming vehicles are detected. It addresses a critical safety gap that affects 210M+ two-wheelers in India alone, at an implementation cost of under ₹200 per vehicle.

- ✅ Eliminates glare hazard: automatically switches beam in <50ms — 30× faster than human reflex
- 💰 Ultra low-cost: LDR + comparator + relay adds <₹200 per motorcycle, enabling mass-market deployment
- 🌍 Scale impact: 210M two-wheelers in India, 21M sold annually — even 10% adoption prevents hundreds of fatalities
- 🔧 Universal retrofit: works with existing H4 halogen and LED headlights — no vehicle modification required
- 🚀 Technology roadmap: LDR → camera detection → AI beam shaping → V2V communication — clear upgrade path

Thank You

Auto-Switching Front Light — AFS for Motorcycles

LDR · AFS · High Beam · Low Beam · Glare Elimination · Two-Wheeler Safety